TICKLE THE PIG · MUD WAR REDESIGN

Songs of the Bog

Build the mire, deploy the horde, hold the line to the beat.

A Guitar-Hero rhythm evolution of the clan war. The skill core becomes a tap-on-the-beat goblin defense; the strategy layer becomes a two-phase, double-blind duel over three snout-rich areas — built on the Fronts substrate that already ships, so the validated coordination math and anti-cheat carry over intact.

7-day war · 2 phases | 3 contested areas | hidden goblin deployment | rhythm defense | flag-gated, off by default

WHAT THIS IS

The idea, in one breath

Two Sounders (5-member clans) fight a 7-day war over **three shared areas** — the Truffle Field, the Bog Bridge, and the Village Gate. First the crew **tends** the areas together (co-op), then each side secretly **deploys** a goblin horde at the other's areas, and finally everyone **holds the line** by playing a short rhythm run: tap on the beat to fling mud and keep the goblins out. Hold more areas than they do, and the rope swings your way.

What changed from "Fronts of the Bog"

- **Skill core → rhythm.** The single-goblin "hold & release" timing toss becomes a short Guitar-Hero stream of goblins you tap on the beat.
- **Fronts → areas you defend.** The 3 abstract contested fronts become 3 areas you actively defend against a deployed wave.
- **Hidden pressure → hidden deployment.** The fuzzy hold-pressure becomes an *active* double-blind: you choose which difficulty wave (easy/med/hard) attacks each of *their* areas.
- **One global grind → two phases.** A co-op **Tend** phase then a skill **Hold** phase.

What stayed (and why that's the point)

- The **multi-contributor fold** `SUM(LEAST(mud,12))` — the super-additive engine the Monte-Carlo sim already certified at **96%** coordination-beats-scatter.
- The **band-enum wire contract** and server-owned scoring (anti-cheat).
- The **fog RLS**, the daily-tug rope, the **Elo ladder**, payouts, titles, cosmetics.
- The **goblin art** (grunt / scout / brute / warboss + hit poses + mud splats) and all of Slop Toss's juice.

> **The single most important design decision.** The rhythm run does *not* replace the team math — it *feeds* it. Two or three crewmates each play a **short** run into the **same** area, and their banked mud sums under the existing 12-cap and concede floor. So the proven "you can't hold an area solo; convergence multiplies" engine stays exactly where it was measured, and the new minigame just becomes the input.

CADENCE

The war loop

Seven days, split across the existing 7-day fronts clock. The 5 "Hold" days map 1:1 onto the 5 already-scored fronts days, so the rope and fold math are untouched.

DAYS 1–2

Tend the Mire

The whole crew slings its flat 7 throws/day (the shipped timing toss) to build depth at the 3 areas. Decide **where to stack**. Self-assign which area you'll hold. No snouts spent.

END OF DAY 2

Loose the Horde

Each crew secretly maps its three waves — **easy** / **med** / **hard** — onto the opponent's three areas. Double-blind: you can't see where their hard wave lands. Re-choosable each Hold day.

DAYS 3–7

Hold the Line

Defenders play short rhythm runs against the wave the enemy deployed. 2–3 can co-defend one area; their mud sums. Each day the rope swings by the margin of areas net-held. Leader has one redeploy token.

DAY 7 (OR ROUT)

The Mire Settles

Rope-side wins; an early rout fires at ±12. Winners split the pot, get the 72h buff, titles, a war-exclusive cosmetic, and the Elo bump. A 24h cooldown blocks instant re-challenge.

THE RUBRIC

Does it satisfy the three pillars?

The same success test the Fronts design was held to — now **sim-validated**. `scripts/sim_deploy.py` Monte-Carlo'd the deploy-Blotto across 5 seeds; all three pass.

> **P0 sim results (5 seeds).** Genuine dilemma — **0 pure saddle points**, no dominant deploy, no dominant defense, best-response cycle ≥2, deploy swings held-V by ~2.7. **coord > scatter 78–80%** (coordination amplifies). **skilled-scatter > unskilled-coord 86–93%** (skill can't be bought with coordination). Mirror draw ~40% (not perpetual). A 3-active crew's maximin defense holds 5V — never auto-locked-out. **The numbers are locked.**

1 · Strategy-predominant pass

Outcome = areas net-won, never raw activity. Three stacked reads vs the opponent: **where to build deep**, **who co-defends what**, and **where each difficulty wave attacks** (the double-blind Blotto).

Sim-confirmed: the deploy×defense matrix has **0 saddle points** and no dominant line; the attacker's best *hard* target is non-constant across where the defender hides weak players. Deploying hard pays off only via hold-pressure on a thin/weak area — never universally.

2 · Coordination amplifies pass

Super-additivity is the *original* fold `SUM(LEAST(mud,12))` across 2+ members above the concede floor — reused, not reinvented. One capped contributor sits below the marquee floor, so holding a contested area **requires** convergence.

Sim-confirmed: **coord > scatter 78–80%** at equal skill. The unvalidated ×1.75 "backing" multiplier was dropped; coordination comes from concentration (the proven (3,2,0) move), and tickle/barn fuel is access-only.

3 · Normalized floor + skill ceiling pass

Floor is dead flat: **no snout buy-in**, the rope moves only on normalized 0–3 band-mud (never wealth/VIP-skewed snouts), flat budgets for everyone. Skill raises the ceiling; a forged perfect run equals an honest flawless one.

Sim-confirmed: **skilled-scatter beats unskilled-coord 86–93%** — coordination can't substitute for skill. Build rule: the access token scores the *submitted* run, not best-of-N, so retries buy **roster relief**, never a higher ceiling for the same skill.

SKILL CORE

The rhythm run (Guitar Hero, cozy)

Goblins stream down a single lane toward a fixed strike post. Tap on the beat to fling mud. It's a direct evolution of Slop Toss's single-strike-zone timing — one thumb, playable muted via the existing gold-post telegraph — so the whole juice stack ports over.

How it plays

Short & retryable. A run is ~12s with **3 scored notes** (plus unscored feint goblins for density). Bands submit *per note*, so a dropped connection loses one note, not the run.

One thumb, temporal window. `classify()` is re-pointed from spatial `|x−0.5|` to temporal `|dt|`: ≤45ms = **perfect**, ≤90ms = **good**, ≤150ms = **weak**, beyond / no tap = **leak**.

Leak = forgone gain, softened. A leak breaks the streak ("he slipped past!") and burns *cosmetic* snouts only. It never steals real spendable snouts and never drags the rope below an honest run.

Why the rope can't be gamed

A hit's value is normalized. Every hit banks the 0–3 band into `mud_front_pushes`; a perfect is always 3, so a skilled defender banks full mud on *any* difficulty (the skill ceiling is untouchable). Difficulty's rope effect is **hold-pressure** — a hard wave raises the bar to hold the area (forcing a 3rd body) and mildly tightens windows (suppressing a *weak* defender). The +2/+3/+4 snout numbers are separate cosmetic juice.

Reused juice. Hit-stop, screenshake, recoil, mud splats, floaters, gold-burst rings, boss vignette, haptics, archetype art, focus-gated lifecycle — all verbatim from `SlopToss.tsx`.

New visually. Multiple charted goblins on the lane at staggered offsets, plus a fixed strike line instead of one moving release.

The three "songs" = three difficulties

SONG	TEMPO / DENSITY	COSMETIC VALUE · LEAK	WHO IT PUNISHES
Bog Shuffle · easy	slow, ~1 goblin on screen, wide feel	+2 · −4	Safe. Low reward, low risk.
Mire March · med	medium, ~2 on screen, scouts mixed in	+3 · −6	The honest default.
Warboss Stomp · hard	fast, ~3 on screen, syncopated, warboss capstone	+4 · −8	A skilled defender nets more; a weak one bleeds.

These numbers touch only the cosmetic snout coffer — the rope always moves on the normalized band-mud.

STRATEGY

Three areas, variable concentration, hidden deployment

Defense: concentration is the lever

Each member self-assigns which area they hold (`mud_war_plans`, reused as-is — **multiple members may pick the same area**). The fold already sums `LEAST(mud,12)` across all committers, so 2–3 defenders on the marquee clear the **0.6·P concede floor** that one alone cannot.

That re-instates the proven **(3, 2, 0)** Blotto move: stack the V=5 marquee, hold the V=4, *concede* the V=3. Spreading 1-1-1 holds nothing.

Offense: the double-blind horde

Each crew secretly assigns its three waves **easy / med / hard** one-to-one onto the opponent's three areas (new `mud_war_deploys`, leader-only, fogged). You point your hard wave at *their* pile without seeing where theirs lands on yours.

Re-choosable per Hold day, so the read → counter → adapt loop survives all the duel rounds. Both crews attack **and** defend, simultaneously, under fog.

> **The leader's actual decision tree.** "Who's our cleanest rhythm player? Put two on the marquee. Where will *they* point hard — at our fat Truffle Field, or will they expect us to stack it and feint hard at the Gate? Do we deploy hard at their weakest defender, or play med everywhere and bank the safe margin?" That tension — not twitch alone — is what decides the war.

No headcount lockout

An area with no submitted run scores a **deterministic floor-defense** (a small fixed effMud from the war seed), not auto-concede — so an absent member degrades the team gracefully. A 3-active crew reliably holds one area; access tokens let them reach for a third.

Bot war is a real trainer

The Mudlarks deploy a deterministic wave per area and defend via the existing `bot_front_eff` scalar. The default onboarding entry point (`challenge_house`) practices the exact mechanic, at zero wallet risk.

ECONOMY

The firewall: snouts as flavor, mud as the verdict

Your idea framed goblins as "stealing snouts." We kept that *feel* but moved the real win condition off the wallet, because the critics showed that staking real snouts breaks Pillar 3 (pay-to-win) and Pillar 6 (cozy loss-aversion). The resolution:

The rope moves only on normalized band-mud. A separate, purely **cosmetic** snout coffer per area is fed by difficulty-scaled hits and drained by leaks (clamped at 0) — pure juice for the "goblins steal snouts" feel, *never* read by the fold. The deploy's real bite on the rope is **hold-pressure** (a hard wave forces more bodies onto an area), not the coffer — so a deep pile can never buffer weak defense, and no wallet is ever at risk.

No buy-in

The Tend phase costs no snouts. The stake is the **mud you earn in-war**, exactly as today — wealth and VIP buy nothing.

Unchanged payout

`resolve_war` keeps its slings-based pot, the `HOUSE_BONUS=25` bot path, the 72h buff, titles, cosmetics and gated Elo. No escrow, no balance debit inside the safety-critical block.

Win-trade cooldown

A new `crews.next_war_at = now()+24h` stamped at resolve blocks a colluding pair from instantly re-challenging to farm the mint.

TEAMWORK FUEL

Tickles & barns — one access lever, kept flat

Your "send tickles / visit barns to give someone the chance to play" instinct is the access economy. We made it **one** lever (not three), so it stays simple and can't leak pay-to-win.

Access fuel = an extra attempt

A teammate's barn-visit or tickle (the unchanged `tickle_at_barn`) mints a flat, war-scoped **access token**: at most **+1 short-song attempt per recipient per Hold day**. It turns a 3-active crew into a 4-active one — **roster relief**, never a per-hit power-up.

It scores the *submitted* run, not best-of-N. So retries buy coverage, not a fake ceiling.

Why it can't be farmed

- **Crew + war scoped**, not friend-graph scoped — an alt ring can't mint tokens by self-friending; it must occupy one of the 5 seats.
- **Flat per-recipient cap** regardless of the sender's (faster-regen VIP) tickle bank — 10 alts grant the same as 1 teammate.
- **Self-leak can't grief.** The fold *sums*, never subtracts: a bad actor tapping all-whiff contributes 0, they cannot drain mud others banked. The leader's redeploy token pulls them.

TUNING

Locked numbers

CONSTANT	VALUE	WHY
`BUILD_DAYS / WAR_DAYS`	2 / 5	Sum to the locked 7-day fronts clock; the 5 Hold days map 1:1 onto the already-scored fronts days so the rope math is untouched.
`AREAS / V`	3 / [5, 4, 3]	The three areas are the three seeded fronts re-skinned (Truffle / Bridge / Gate) — inherits the validated empty-dominant-set Blotto with zero schema growth.
`NOTES_PER_RUN`	3 scored	Short, so 2–3 members co-defend one area and their mud SUMS under the 12-cap. A clean run banks ~9; two committers clear the marquee floor (~16) one can't.
`RUN_LENGTH_MS`	12000	A glanceable, retryable run for a single distracted mobile sitting — not a 32s one-shot a notification can destroy.
`RUNS_PER_DAY (Hold)`	2 + up to 1 token	Replaces the 7/21 throw budget on Hold days. Two flat runs for everyone; the access token grants at most one extra.
`PERFECT / GOOD / WEAK`	45 / 90 / 150 ms	Temporal analogs of Slop Toss's gold/green/weak half-widths; tightens to 35ms on hard via the existing streak ramp.
`HOLD_COEF`	0.78	**Sim-locked.** Hold a contested area iff banked mud ≥ `HOLD_COEF · P · PRESS[diff]`. Tuned so a *hard* marquee needs 3 bodies (2×cap=24 < 0.78·26·1.30=26.4) while med needs 2 — the lever that makes concentration pay.

`PRESS (hold-pressure)`	easy 0.80 · med 1.0 · hard 1.30	**Sim-locked.** The deploy's rope effect: a harder wave raises the bar to hold the area, forcing more defenders. This is what makes the hidden deployment matter to the outcome.
`window mult / skill σ`	1.30/1.0/0.82 · 34/64/108ms	Difficulty also mildly tightens windows (suppresses a weak defender's mud); σ is the per-skill timing-error calibration (high banks ~full on any difficulty → skill ceiling preserved).
`coffer value / leak`	e +2/−4 · m +3/−6 · h +4/−8	**Cosmetic only** (off-rope juice for the "stolen snouts" feel). Free to tune — never touches the win condition.
`PER_AREA_CAP`	12	Reused PER_FRONT_CAP — caps one member's mud per area so holding still needs 2+ real defenders (the lone-ace kill, now load-bearing on defense).
`FRONT_SCALE / clamps / ROUT`	4 / ±4,±2,±5 / 12	Unchanged rope math; the rhythm outcome feeds `coord_notch` through the exact fold the sim certified.
`ATTEMPT_TOKEN_CAP / RALLY`	1 / 1	Flat per-recipient and per-pair caps sever VIP tickle-wealth from in-war power and cap alt-ring farming.
`WAR_COOLDOWN_HOURS`	24	Kills back-to-back win-trade cycling of the resolve mint.
`CHART_SEED`	`hashtext(war:day:area:diff:caller) & 2147483647`	Reuses seed_war_board's masked-hash pattern; folding caller_id in defeats the copy-a-perfect-array exploit between co-defenders.

`flag default`	mud_rhythm_on() = false	Same immutable-false gate as mud_fronts_on(); off-path reproduces migration 20260667 byte-for-byte.

IMPLEMENTATION SURFACE

Data model & anti-cheat

One migration — `supabase/migrations/20260668000000_mud_war_rhythm.sql` — carrying every touched function verbatim once (no carry-latest clobber), flag-gated off, with a golden-output regression test.

`reuse` **as-is**

- `mud_wars` lifecycle + sweep
- `mud_front_pushes` — the super-additive fold, for both phases
- `mud_war_plans` — self-assign, multi-per-area
- `mud_war_fronts` board (V=[5,4,3])
- `resolve_war`, `score_mud_war_days`, `fold_front_outcome/margin`, `seed_war_board`
- `crew_ratings` + Elo + `/clan-ladder`
- fog RLS + `test_fog_rls.sql`
- `tickle_at_barn` + `barn_visits`
- Slop Toss juice + FrontBoard UI + goblin art

`change` **additively**

- `mud_wars` += `build_ends_at`, `next_war_at`, `rhythm_enabled`
- `mud_war_fronts` += `snout_coffer` (cosmetic; never read by the rope)
- `score_mud_war_days` phase-gated — no-op on Tend days, unchanged on Hold
- `throw_mud` verbatim + sibling `submit_run` (band array)
- `challenge_*` += cooldown check & stamp
- `war_state` += phase / deploy / snoutCoffer / accessTokens (superset; old clients ignore)

`new` **tables**

`mud_war_deploys`

(war, day, crew, target_area, difficulty)

The hidden double-blind. Inherits the fronts fog RLS + realtime exclusion.

`mud_war_access`

(war, recipient, day, tokens)

War-scoped access pool minted from barn visits. The only place tickles touch the war.

Rhythm anti-cheat, end to end

The wire contract stays **band-enum-only**. A Hold run submits an **array** of up to 3 bands. The server:

1. regenerates the deterministic chart from `CHART_SEED` to learn note count + per-note archetype;
2. validates `len(bands) == chart length` (reject mismatch — anti-replay; pad missing as leaks, truncate overlong);
3. maps each band via the **existing** `throw_mud` CASE (whiff/weak/good/perfect → 0/1/2/3), clamps per note;
4. banks the summed normalized mud into `mud_front_pushes` and applies the cosmetic coffer delta;
5. enforces the per-day run budget via the same race-safe `ON CONFLICT` guard.

Folding `caller_id` into the seed defeats the copy-a-perfect-array exploit between co-defenders. A forged all-perfect array yields exactly an honest flawless run — *exploit ceiling equals skill ceiling*. The fog invariant (opponent's deployment unreadable until reveal) gets a dedicated regression assertion in `test_fog_rls.sql`.

DECISIONS PENDING

Open forks

Proceeding on the bolded default; redirect any time.

QUESTION	DEFAULT	ALTERNATIVE
How many may co-defend one area?	Uncapped headcount, each capped at 12 mud — so concentration helps via more committers and (3,2,0) stays live.	Hard-cap 2/area to force spread — only if the sim shows 3-on-marquee trivially solves it.
Does the cosmetic coffer pay out?	Pure juice / vanity — never pays real snouts, keeps the firewall absolute.	A tiny house-funded bonus for a full coffer — reintroduces a wealth-adjacent term; defer past v1.
Deploy authority?	Leader-only (matches the redeploy gate; clean single-caller double-blind).	Each member deploys their own counter-wave — richer but multiplies the fog surface; defer.
Access-token timing?	Same-day extra run on an under-defended area (legible relief).	Banked for the next Hold day — enables an alpha-strike; snowball risk; defer.
Difficulty rope knobs final?	**RESOLVED.** `PRESS=0.80/1.0/1.30`, `HOLD_COEF=0.78`, window mult 1.30/1.0/0.82 — locked from `sim_deploy.py` (genuine dilemma across 5 seeds).	The cosmetic coffer numbers stay free to tune; a wider parameter sweep before prod is still wise.

ROADMAP

Implementation plan

P0✓

Sim-gate the deploy Blotto *before any code* — DONE, PASS

`scripts/sim_deploy.py` Monte-Carlo'd the deploy game across 5 seeds. It **refuted** the synthesis's "difficulty is cosmetic-only" claim (a cosmetic difficulty can't move the rope, so the Blotto would be flavour) and **found the fix**: difficulty raises the area's **hold-pressure**. Result: genuine dilemma (0 saddle points, no dominant deploy/defense, BR cycle ≥2), **coord>scatter 78–80%**, **skilled-scatter>unskilled-coord 86–93%**, mirror draw ~40%, 3-active holds 5V. Numbers locked.

`scripts/sim_deploy.py · scripts/sim_fronts.py`

Gate passed. Difficulty rope knobs (PRESS / HOLD_COEF / window mult) locked from the sim.

P1

One migration: schema + anti-cheat + folds, flag-gated off

`20260668000000_mud_war_rhythm.sql` carrying every touched function verbatim once: the new columns + `mud_war_deploys` / `mud_war_access` with fog RLS; the `submit_run` RPC; `set_deploy` (leader-only); the phase gate in `score_mud_war_days`; the cooldown in `challenge_*`; bot deploy/defense; `mud_rhythm_on()=false`. Plus a golden-output test asserting byte-identical behavior with the flag off, and two new fog assertions.

`supabase/migrations/20260668000000_mud_war_rhythm.sql · scripts/test_fog_rls.sql · scripts/golden_output.sql`

Carry-latest footgun (build 93). Mitigation: single migration, golden regression, diff every carried body against 20260667.

P2

Constants + client RPC wrappers + hook

`constants/mudFights.ts` gets the locked numbers (client mirror). `utils/mudWars.ts` gets `submitRun` / `setDeploy` wrappers + extended types (phase, deploy, snoutCoffer, accessTokens). `hooks/useMudWar.ts` becomes phase-aware with an optimistic band-mud bump reusing the throwBand pattern; deploys stay *off* realtime (fog).

`constants/mudFights.ts · utils/mudWars.ts · hooks/useMudWar.ts`

Client/server constant drift — keep the existing MUST-match comment block; the golden test pins the server side.

P3

Rhythm conveyor in the minigame component

Rewrite the lap loop in `SlopToss.tsx` (or a new `RhythmDefense.tsx`): a server-charted note conveyor, `classify()` re-pointed to temporal `|dt|`, a per-note tap handler accumulating `bands[]`, incremental per-note submit, and an `onRunComplete(area, bands[])` emit. Reuse the entire `fire()` juice body, archetypes, telegraph, haptics; soften the leak copy.

`components/mudwar/SlopToss.tsx · components/mudwar/RhythmDefense.tsx (optional)`

Juice regressions during the lap→conveyor rewrite; keep classify/scorePoints/fire signatures stable and unit-test `classify(|dt|)`.

P4

Two-phase board UI + deploy / hold sheets

`FrontBoard.tsx` renders Tend vs Hold, area depth, my deployed waves (own live), and the post-fold recap (both sides + which difficulty hit where). `app/mud-war.tsx` gets the phase-aware layout — Tend shows the toss + assignment; Hold shows the rhythm run + a deploy sheet + an access indicator. Ship the living spec + this HTML.

components/mudwar/FrontBoard.tsx · app/mud-war.tsx · docs/mudwar-rhythm-design.md

UI complexity vs "deeper yet simple" — verify the 3-line onboarding ("Tend 3 areas, then hold them to the beat; deploy your hardest wave where they're weakest") is actually true before shipping.

P5

Access fuel + abuse hardening + cooldown verify

A `barn_visits` post-hook mints crew+war-scoped access tokens (1/pair/day, 1/recipient/day) during a Hold-phase war. Verify the cooldown blocks back-to-back challenges, self-leak can't drain others' mud, and an offline area scores floor-defense (not auto-concede).

supabase/migrations/20260668000000_mud_war_rhythm.sql · hooks/useMudWar.ts · utils/mudWars.ts

Alt-ring token farming if minting leaks to the friends graph — gate strictly on is_crew_member / is_war_participant.

HONESTY

Residual risks & mitigations

RISK	MITIGATION
The difficulty-relative leak numbers are *asserted*, not yet proven to yield an empty dominant set — if hard still dominates, Pillar 1 regresses to "point hard at the juiciest pile."	P0 hard gate: `sim_deploy.py` must show non-constant best-difficulty + a best-response cycle before any migration code. Tune in-sim only.
Co-defense with short runs might let 3-on-marquee trivially solve the marquee, collapsing the concentration dilemma into "always stack the V=5."	Re-run the coord-vs-scatter sim with the summed-short-run model; if auto-won, raise the marquee P or cap committers/area (open fork) until (3,2,0)-vs-spread stays live.
Four-migration carry-latest footgun (build 93 already shipped this once) — silently deleting features from 20260667.	ONE migration carrying every touched body verbatim; a golden-output regression asserting byte-identical behavior with the flag off; diff each carried body before push.
Inherited resolve mint + win-trading still leaks value even with the cooldown.	The 24h cooldown caps cycle frequency and the rope now moves only on normalized skill-mud (no escrow to extract). If telemetry shows inflation, add a flat house-funded sink in a *separate* later migration, never inside the fold.
Offline / under-roster crews lose areas they can't physically cover across 5 Hold days.	Deterministic floor-defense for un-played areas; access tokens reach a third area; the concede floor gives graceful degradation (3-active reliably holds 1).
Cozy-fit: a rhythm-accuracy duel can read as sweaty against the gentle voice.	Short retryable runs (12s, per-note submit, dropped runs cost nothing), wide default windows, muted-playable, leak = "he slipped past!" not theft, all failure copy softened.

TO ACTUALLY RUN IT

Activation checklist

1. **Pass P0.** Build and green `scripts/sim_deploy.py`; lock the difficulty numbers from its output.
2. **Author + audit** the single migration; run the golden-output + fog regression tests.
3. **Apply the migration** locally — requires an explicit user "go" (never auto-pushed).

4. **Flip** `mud_rhythm_on()` → true to activate rhythm wars on new challenges.
5. **Playtest** in dev via the `__DEV__` fast-forward harness: verify Tend banks mud, Deploy fogs, a Hold run commits per-note + sums across co-defenders, the rope folds, the recap reveals both sides, a rout resolves + updates the ladder, and the off-path (flag off) is byte-identical.

Songs of the Bog — a fork of the Fronts of the Bog clan layer toward a Guitar-Hero rhythm core, two-phase co-op→duel cadence, and an active double-blind goblin deployment. Designed to make all three pillars true while honoring "deeper yet relatively simple," and to reuse the validated Fronts substrate (the super-additive fold, the band-enum anti-cheat, the fog RLS, the Elo ladder) rather than rebuild it.

Synthesized from a 12-agent design loop — 5 dimension drafts, 6 adversarial pillar/ops/exploit/feel critics (12 blockers, all folded in), 1 synthesis. Companion living doc: `docs/mudwar-rhythm-design.md`.